

TransCMFD: An adaptive transformer for copy-move forgery detection

Enji Liang^a, Kuiyuan Zhang^a, Zhongyun Hua^a, Yuanman Li^b, Xiaohua Jia^{a,c}

^a School of Computer Science and Technology, Harbin Institute of Technology, Shenzhen, Guangdong 518055, China

^b College of Electronics and Information Engineering, Shenzhen University, Shenzhen 518060, China

^c Department of Computer Science, City University of Hong Kong, Hong Kong Special Administrative Region of China

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ABSTRACT

Copy-move forgery is one of the most common image tampering methods. It is a frequently employed method for manipulating evidence or deceiving the public by hiding some objects in an image or replicating significant objects. Therefore, it is crucial to focus on copy-move forgery detection. In this paper, we propose TransCMFD as a new transformer-based model for detecting copy-move forgery. We propose an adaptive transformer encoder and combine the traditional convolution encoder-decoder to capture different global and local features of the forgery image, respectively. This can enhance the model's comprehension of the impact of forgery across different tampered image regions. To allow the model to concentrate more on the tampered regions that resemble the original regions, we introduce a similarity detection module. Moreover, to enhance the localization accuracy of the tampered regions, we design an adaptive loss function combination strategy that incorporates the DICE coefficient loss and binary cross-entropy loss. We perform comprehensive experiments on both synthetic and four publicly available datasets. The results show that our model has better performance in copy-move forgery detection compared to baseline methods, and it has remarkable robustness to some common image attacks such as noise addition attacks, image blurring attacks, and color reduction attacks.

1. Introduction

Nowadays, with the popularity of digital equipment for acquiring images, it has become more convenient to use digital photos as evidence. However, digital photos can be easily edited using sophisticated image editing software. While photo editing software can increase enjoyment, it also offers attackers the opportunity to maliciously manipulate photos [1,2], compromising digital security [3,4]. Therefore, it is imperative to determine whether an image has been manipulated.

Image manipulation involves various methods, including but not limited to image splicing [5], image removal, and copy-move forgery [6–10]. Among various image manipulation methods, copy-move forgery is extensively employed and straightforward methods [11,12]. This manipulation technique copies a segment of an image and makes the copied part look authentic through image processing technology such as scaling, rotation, and color harmony. The copied area is then added into a different part, hiding specific elements and introducing content that was not present in the original image [13–15]. This type of forgery can significantly impact the authenticity and reliability of the content. Furthermore, the application of post-processing techniques can produce a more natural and convincing appearance of the manipulated region, which poses a greater challenge for the detection of manipulated areas.

It is very important to maintain the integrity of visual content to detect copy-move forgery. Hence, it is crucial to devise robust and reliable techniques for detecting image copy-move forgery [11,16].

Generally, the forgery detection method is divided into traditional methods and deep learning-based methods. In the early research process, many researchers focused on using traditional methods for tamper detection. Traditional copy-move forgery detection methods were developed based on hand-designed features and can be divided into two types: block-based methods and keypoint-based methods. These types of methods complete some common steps such as image preprocessing, feature extraction and matching, post-processing, and completing forgery detection. Block-based methods initially segment the image into non-overlapping blocks, which are then processed using feature extraction approaches such as principal component analysis (PCA) [17] and discrete wavelet transform (DWT) [18]. These methods aim to identify similar image patches and locate the areas that may be manipulated. The keypoint-based methods contain three main stages: identifying feature key points of the suspicious image, extracting useful features from these key points, and detecting the forgery area by matching the extracted features. Two commonly used techniques for keypoint feature

* Corresponding author.

E-mail addresses: lajihuya@163.com (E. Liang), zkyhitsz@gmail.com (K. Zhang), huazhongyun@hit.edu.cn (Z. Hua), yuanmanli@szu.edu.cn (Y. Li), csjia@cityu.edu.hk (X. Jia).

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extraction are scale-invariant feature transform (SIFT) [19] and speed-up robust feature (SURF) [20]. The keypoint-based detection methods have a high detection failure rate when dealing with smooth tampered regions. Additionally, these conventional techniques mainly depend on manual extraction features, leading to limited accuracy and robustness. The effectiveness of these traditional methods is often affected by factors such as parameter settings and algorithm selection, resulting in high computing costs and difficulty in meeting real-time requirements.

Recently, deep learning techniques such as convolutional neural network (CNN) have shown remarkable achievements in image forensics [21–25]. Among the various topics of image manipulation detection, using deep learning for forgery detection has received widespread attention [26–28]. Wu et al. [27] proposed a model using CNN to detect forged regions and calculate correlation maps in convolutional feature blocks. Furthermore, it uses a deconvolutional network to reconstruct the detected forgery masks. Chen et al. [13] enhanced the source/target region discrimination in the similarity detection and operation localization branches of Buster-Net to achieve improved performance. Despite the advancements, current methods primarily focus on either traditional approaches or CNN-based techniques, each with its limitations. Traditional methods rely heavily on manually designed features and struggle with complex tampered regions. Although CNN methods do not require manual feature design, often ignore global features and primarily focus on local features. To address these limitations, this paper explores integrating transformer-based approaches with CNNs for copy-move forgery detection. The key research question we aim to address is: can integrating transformer with CNNs enhance the detection of copy-move forgery by capturing both local features and global context, thereby improving accuracy and robustness?

In this work, we propose and evaluate an innovative copy-move forgery detection model called TransCMFD. To enhance the model's ability to capture effective features and global correlations, we designed a novel transformer encoder called the adaptive transformer module. This module allows the model to precisely localize the tampered regions. By integrating the adaptive transformer encoder into CNN, our model can capture effective context features. Additionally, to further improve the detection performance, we introduce a feature similarity calculation module that focuses specifically on tampered regions. Our proposed TransCMFD aims to bridge the gap between traditional methods and modern deep learning approaches by leveraging the strengths of both, providing a more robust and accurate method for copy-move forgery detection.

The summary of this work's primary contributions is as follows:

- We propose TransCMFD as a novel model that employs an adaptive transformer to capture global context information and accurately locate the tampered regions. To further enhance the detection ability of tampered regions, we introduce an adaptive loss function combination strategy.
- We introduce a feature similarity module that effectively utilizes the useful feature relationships of similar image patches in the copy-move forgery image.
- We conduct comprehensive experiments on four selected benchmarks, and the obtained evaluation demonstrates that our model performs better than the baseline models while maintaining high robustness against post-processing attacks.

The subsequent sections of this paper are structured as follows. Section 2 explores some related methods of forgery detection. Section 3 illustrates the details of our copy-move forgery detection model. Section 4 describes our experiments on different datasets and verifies our network's robustness under various attacks. We conclude this paper in Section 5.

2. Related work

In this section, we introduce prior copy-move forgery detection approaches and some background knowledge.

2.1. Image forgery detection

To detect various types of image forgery, numerous deep learning-based techniques have recently been developed [29–33]. Bunk et al. [34] introduced a forgery detection model with a patch classifier based on deep neural networks. The model generates a heatmap using CNN and a Gaussian conditional random field model. Subsequently, a random walk segmentation approach is employed for identifying manipulated regions. Some methods [35,36] have been developed to learn hierarchical representations from the input color images to deal with RGB color images. Rao et al. [35] proposed a CNN model with manually constructed modules. However, in this model, the pre-training of the dense fully convolutional networks may be hindered by the inherent differences in forgery types. This results in a significant reliance on the distribution of data types. To overcome this challenge, Zhong et al. [37] proposed Dense-InceptionNet that uses the dense connection to extract multiscale features more efficiently to detect manipulation regions. Zhuang et al. [38] proposed a densely connected and dilated convolutional encoder–decoder architecture to detect the manipulation area. Nevertheless, the model has limited scalability because it lacks a suitable module and can only be applied to pre-trained models.

In addition, some forgery detection methods aim to distinguish the forged regions and target regions [13,26,39,40]. For example, Islam et al. [40] introduced a unified model with two-stage attention, using location information and pixel-to-pixel interdependence. This method uses the Generative Adversarial Network (GAN) [41] to differentiate between the source and target regions. However, common deep learning-based models often rely solely on CNNs to extract local features while ignoring global features. Since an image usually carries complete meaning through contextual information, the local features are insufficient to identify the manipulated regions. To address this issue, global features of the entire image should be utilized for forgery detection. Our model introduces an adaptive transformer to help capture useful tampered area features and locate the tampered areas.

2.2. Attention mechanism

In the exploration of deep learning networks, some scholars have proposed the attention mechanism [42–44]. The mechanism aims to simulate human visual attention and can help reduce computational complexity in image processing tasks by selectively focusing on specific areas of an image [45]. Furthermore, most attention mechanisms can be jointly trained with the underlying deep learning methods such as CNN with conventional backpropagation. The self-attention mechanism [46] is a significant work in attention mechanism models. The self-attention has demonstrated remarkable success in sequence prediction tasks and recently demonstrated remarkable performance in deep learning computer vision [47]. To address the weak ability of CNN networks to extract global features, recent studies have explored the use of the self-attention module to improve performance in image manipulation detection models [48]. These studies demonstrate that attention mechanisms can enhance the robustness of image-processing detection methods.

2.3. Transformer

The transformer [46] is a revolutionary deep neural network architecture mainly based on the self-attention mechanism. Researchers have investigated the integration of transformers to computer vision tasks due to powerful feature expression capabilities. Transformer-based models have demonstrated remarkable performance on various vision benchmarks compared to traditional CNN models. Recently, transformer networks have received increasing attention in image-processing tasks due to high performance and lower dependence of the

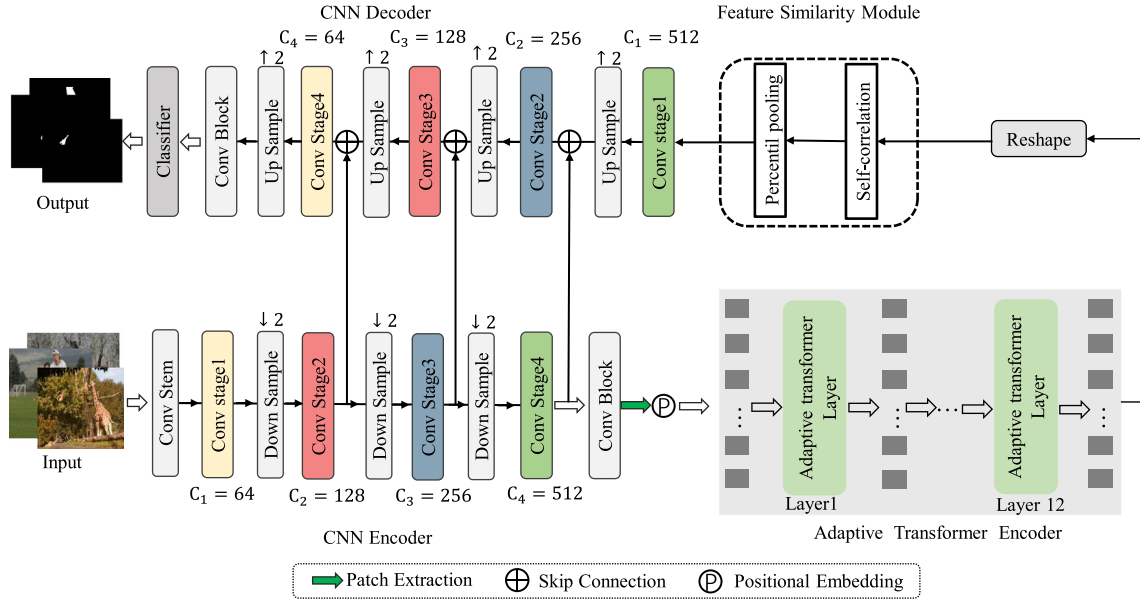


Fig. 1. Framework of our TransCMFD, which consists of CNN encoder-decoder, adaptive transformer encoder, and feature similarity module. The CNN encoder is used to extract the local features of the image; the adaptive transformer encoder extracts global features within each block; the feature similarity module selects similar features in the image and the original area; finally, the CNN decoder up-samples the features and outputs the prediction mask through the classifier.

transformer on vision-specific perceptual distortions. Transformer models have proven particularly effective when used for high- and mid-level vision tasks. For example, the DETR model [49] and the deformable DETR model [50] for object detection. Moreover, transformer networks have shown good performance on other vision tasks. For instance, Liu et al. proposed ConvTransformer [51], a video frame synthesis architecture that not only allows extrapolation of video frames but also interpolates photorealistic mean frames.

While some existing works have demonstrated the effectiveness of transformer networks in forgery detection tasks, these methods have shown certain limitations. For instance, Horvath et al. [52] proposed a transformer model for satellite image forgery detection. However, this method solely relies on a pure vision transformer for unsupervised satellite image forgery detection, resulting in bad performance in tampered area detection and robustness. Sun et al. [53] introduced a novel transformer for tampered region localization. Despite these efforts, traditional transformer-based image forgery detection methods continue to be hindered by their limited sensitivity to the spatial structure of images. As a result, they may not fully capture the spatial information in tampered images and are unable to dynamically adjust attention weights, leading to reduced detection accuracy and poorer performance, particularly in complex manipulation scenarios.

3. The design of TransCMFD

In this section, we present the details of TransCMFD, and Fig. 1 illustrates its structure. Inspired by the transformer architecture, we propose a novel adaptive transformer model, allowing our model to learn the subtle forgery artifacts of a forgery image. Consequently, our model can effectively capture the abundant information of tampered images, thereby enhancing the accuracy and robustness of detecting invisible forged artifacts. As shown in Fig. 1, the novel model includes the CNN encoder module, adaptive transformer encoder module, feature similarity module, and CNN decoder. We consider the input image $\mathcal{X}$ with dimensions $H \times W \times C$, where H , W , and C represent the height, width, and number of channels, respectively.

3.1. CNN encode-decoder

3.1.1. Encoder

The CNN encoder employs a hierarchical method to effectively extract local features of the copy-move forgery image through convolution operations. The process starts with a convolutional stem, which is used to extract the initial local features from the input forgery image $\mathcal{X}$. The convolutional stem includes a size of 7×7 convolution block, a size of 3×3 max-pooling layer, a GroupNorm layer [54], and a ReLU layer. Both convolution layers and max-pooling layers keep strides of 2×2 to down-sample input forgery images. Following the convolutional stem, the CNN encoder further utilizes four CNN stages to learn local features. The i th stage contains R_i residual convolution blocks, similar to the architecture of ResNet50 [55].

3.1.2. Decoder

The decoder aims to restore the original image resolution by employing cascading successive upsampling blocks. Each upsampling block includes a sequence of a $2 \times$ upsampling operator, a 3×3 convolution block, and a ReLU activation. Moreover, we fuse the features from the CNN encoder at the decoder stage to facilitate model training. This fusion of features enables the model to generate high-quality masks with precise boundary delineation, which can greatly help to detect tampered regions.

Feature fusion. In our model, we employ skip connections to integrate features extracted from the CNN encoder with features obtained from the CNN decoder. This fusion is achieved through a feature concatenation layer, which combines fuse feature maps of the encoder-decoder at appropriate positions. The skip connections operation facilitates the fusion of feature maps across different layers and allows the model to leverage rich detail information from the high-level convolutional feature layers. This is achieved by reconstructing features of adjacent layers and reconstructing features across multiple layers, thereby improving and enhancing the transfer of feature information between layers. In this way, the model can reduce feature loss and preserve higher-resolution details during the learning process, improving the accuracy of detecting tampered regions.

Algorithm 1 Adaptive Multi-Head Self-Attention (adaptiveMSA)

Input: Feature map F_p , trainable projection matrices $\{U_Q^i\}, \{U_K^i\}, \{U_V^i\}$ for each $Head_i$, learnable parameters $\{S1, S2\}$, bias vectors $\{b1, b2\}$, weights $\{\epsilon, \delta\}$, trainable projection matrix W_O .

Output: Feature map $adaptiveMSA(F_p)$.

```

1: for  $i$  from 1 to  $h$  do
2:    $Q_i \leftarrow F_p * U_Q^i, K_i \leftarrow F_p * U_K^i, V_i \leftarrow F_p * U_V^i$ .           ▷ Generate query  $Q_i$ , key  $K_i$ , and value  $V_i$  for each head  $i$ .
3:    $d_k \leftarrow \text{dimension of } K_i$ .
4:    $attention\_weights_i \leftarrow Q_i K_i^T / \sqrt{d_k}$ .                               ▷ Compute attention weights.
5:    $attention\_probs_i \leftarrow \text{softmax}(attention\_weights_i)$ .                     ▷ Normalize the attention scores to obtain probabilities.
6:    $Head_i \leftarrow attention\_probs_i * V_i$ .                                       ▷ Calculate the attention output for head  $i$ .
7:    $M1\_Head_i \leftarrow S1 * Head_i + b1$ .                                         ▷ Map the attention heads through linear transformations with learnable parameters and biases.
8:    $M2\_Head_i \leftarrow S2 * Head_i + b2$ .                                         ▷ Map the attention heads through linear transformations with learnable parameters and biases.
9:    $Head_i \leftarrow \epsilon * M1\_Head_i + \delta * M2\_Head_i$ .                             ▷ Combine the mapped heads with the given weights.
10: end for
11:  $concatenated\_Heads \leftarrow \text{Concat}(Head_1, \dots, Head_h)$ .                     ▷ Concatenate all heads.
12:  $adaptiveMSA(F_p) \leftarrow concatenated\_Heads * W_O$ .                         ▷ Project the concatenated Heads with a trainable projection matrix  $W_O$ .
```

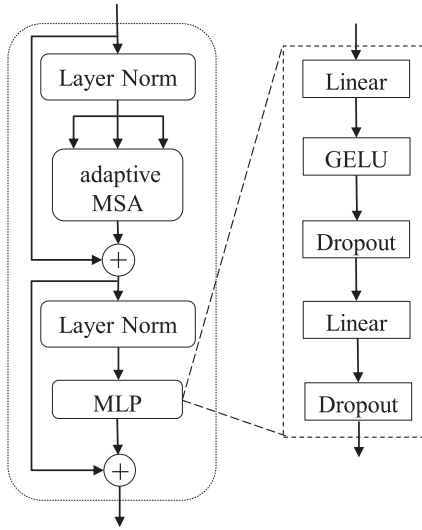


Fig. 2. The adaptive transformer layer and MLP layer internal structure diagram.

3.2. Adaptive transformer encoder

The novel transformer encoder is proposed to adaptively learn the global representation of the tampered image. This encoder is composed of multiple adaptive transformer layers and each adaptive transformer layer incorporates an adaptive multi-head self-attention block (adaptiveMSA), where the adaptiveMSA is designed to assign different attention weights to each input vector. As shown in Fig. 2, the calculation of the c th encoder layer is represented as follows:

$$\begin{aligned} \mathbf{r}'_c &= \text{adaptiveMSA}(\mathbb{L}(\mathbf{r}_{c-1})) + \mathbf{r}_{c-1}, \\ \mathbf{r}_c &= \mathbb{M}(\mathbb{L}(\mathbf{r}'_c)) + \mathbf{r}'_c, \end{aligned} \quad (1)$$

where $\mathbb{L}(\cdot)$ denotes the layer normalization [56]. $\mathbb{M}$ denotes MLP operation and $\mathbf{r}_c$ is the encoded representation.

Tokenization and patch embedding. Before entering into the adaptive transformer encoder, the local features extracted by the CNN encoder are tokenized by reshaping it into a series of flattened 2D patches. Assume that each patch is of size $P \times P$, we can obtain the tokenized patches as $\{\mathbf{F}_p^i \in \mathbb{R}^{P^2 \times C} \mid i = 1, \dots, K\}$, where $K = \frac{H \times W}{P^2}$ is the number of image patches. We convert the patches $\mathbf{F}_p$ into a latent embedding space of dimension d by using a linear projection. To preserve positional information, we additionally include a learnable position embedding, represented by Eq. (2):

$$\mathbf{r}_0 = [\mathbf{F}_p^1 \mathbf{E}; \mathbf{F}_p^2 \mathbf{E}; \dots; \mathbf{F}_p^K \mathbf{E}] + \mathbf{E}_{\text{pos}}, \quad (2)$$

where $\mathbf{E} \in \mathbb{R}^{(P^2 \times C) \times d}$ is the patch embedding projection, and $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{K \times n}$ denotes the learnable position embedding.

Adaptive multi-head self-attention. We design the adaptiveMSA by introducing a dual-path adaptive attention mechanism (DPAAM) into the traditional multi-head self-attention (MHSA). When calculating the attention matrix, the adaptive attention module dynamically selects the head used for attention calculation according to the characteristics of the input sequence. Specifically, DPAAM introduces two additional learnable parameters $S1$ and $S2$. The proposed operation adaptively combines information from different positions in the encoder, facilitating richer interactions between positions. This allows the model to capture more complex relationships between the tampered and original regions, improving the accuracy of detection. Selectively combining information at each position improves the feature expression of the input and allows the proposed method to better distinguish between tampered and original regions. The operation enhances the performance of the forgery detection system by effectively modeling the complex dependencies between different regions of the input. We also provide pseudo-code for adaptiveMSA to illustrate the process, as shown in Algorithm 1.

The adaptive attention module is similar to the traditional attention module, but its weight matrix changes dynamically. To calculate the adaptiveMSA in the adaptive transformer, we denote with $\mathbf{F}_p$ the tokenized patches. We need to compute the query Q , key K , and value V by linearly transforming tokenized patches into three parts. Their computation equations are as follows:

$$\begin{aligned} Q &= \mathbf{F}_p U_Q^q, \\ K &= \mathbf{F}_p U_K^k, \\ V &= \mathbf{F}_p U_V^v, \end{aligned} \quad (3)$$

where $U_Q^q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $U_K^k \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $U_V^v \in \mathbb{R}^{d_{\text{model}} \times d_v}$ represent the trainable projection parameter matrix. d_{model} represents the hidden layer dimension. h represents the head number of the MHSA module. We set $d_k = d_v = d_{\text{model}}/h$ in our experiment.

After calculating Q, K, V , we can compute attention weights $Att_i \in \mathbb{R}^{L_q \times L_k}$ as follows:

$$Att_i = \frac{QK^T}{\sqrt{d_k}} \in \mathbb{R}^{L_q \times L_k}, \quad (4)$$

where L_q represents the query sequence length and L_k represents the key sequence length.

We can then initially obtain $Head'_i$ by multiplying the attention weight Att_i by the value V :

$$Head'_i = \text{softmax}(Att_i) V \in \mathbb{R}^{L_q \times L_k}. \quad (5)$$

Then $M1$ and $M2$ are obtained by mapping the results through.

$$M1(Head') = S1 * Head'_i + b_1,$$

$$M2(Head'_i) = S2 \cdot Head'_i + b_2, \quad (6)$$

Where b_1 and b_2 are learnable bias vectors, and $S1$ and $S2$ are learnable parameters.

Finally, we need to weigh $M1$ and $M2$ to obtain the final $Head_i$.

$$Head_i = \epsilon \cdot M1(Head'_i) + \delta \cdot M2(Head'_i), \quad (7)$$

where $\epsilon, \delta \in (0, 1)$ denote the weights for each map. Then the final output can be obtained by concatenating all the h heads and projecting them with a trainable projection matrix $W^O \in \mathbb{R}^{hd_v \times d_{model}}$, namely

$$\text{adaptiveMSA} = \text{Concat}(Head_1, \dots, Head_h) W^O \quad (8)$$

3.3. Feature similarity module

We introduce the feature similarity module (FSM) [26] to help detect the forgery regions. Before the output tokens are fed into the FSM from the transformer encoder, the tokens need to be transformed into an image-like feature F and processed by a self-correlation calculation block and a percentile pooling block.

Self-correlation calculation block. The block adaptively divides the image-like feature B into 16×16 blocks (or windows) $\{B_{r,c} | r, c \in [0, 15]\}$ and then calculates the similarity between all pairs of features blocks.

To measure the degree of association between two feature blocks, we employ the Pearson correlation coefficient, which was first introduced in [57]. Specifically, the Pearson correlation coefficient ranges from -1 to 1 . Assume that two feature blocks are $B[i]$ and $B[j]$, where $i = (i_r, i_c)$ and $j = (j_r, j_c)$. The feature similarity between them, using the Pearson correlation coefficient τ , can be expressed as:

$$\tau(i, j) = (\tilde{B}[i])^T \tilde{B}[j] / 512, \quad (9)$$

where $\tilde{B}$ is the normalized version of a vector B and $\tilde{B} = (B - \mu_B) / \sigma_B$, where μ_B is the mean value and σ_B is the variance. The τ indicates a larger similarity of the two feature blocks. For a $B[i]$, its score vector $S_X[i]$ is defined as follows:

$$S_X[i] = [\tau(i, 0), \tau(i, 1), \dots, \tau(i, j), \dots, \tau(i, 255)]. \quad (10)$$

Percentile pooling block. The percentile pooling block is used to adaptively select the most similar feature blocks for each feature block, and enhance the robustness of our model for the size of input images. After the self-correlation calculation, we can obtain a scoring matrix $S_X \in \mathbb{R}^{256 \times 256}$, where $S_X[i, j]$ denotes the similarity between the feature blocks $B[i]$ and $B[j]$.

The percentile pooling operation first sorts each $S_X[i]$ in descending order and obtain $S'_X[i]$:

$$S'_X[i] = \text{sort}(S_X[i]). \quad (11)$$

The similarity among distinct regions of an image is affected by various factors such as patterns, and textures. Generally, it can be observed that larger pairs of regions tend to exhibit a higher level of similarity. In the block responsible for calculating self-correlation, the similarity between image blocks is computed, and the distribution of similarity values is dependent on the size of the block. However, to mitigate the impact of varying similarity distributions across different sizes of tampered input images, we employ percentile pooling. This technique allows the model to focus on the top p -percentage of the most significant similarity scores. In this module, K scores need to be selected to form a combined percentile score vector $P_X[i]$. More specifically, the filtering process is illustrated as follows:

$$P_X[i][k] = \begin{cases} S'_X[i][T], & 0 \leq k < T \\ 0, & k \geq T \end{cases} \quad (12)$$

where T is the index after mapping the percentile $p_k \in [0, 1]$ to the original sorted score vector. We can get T as follows:

$$T = \text{round}(p_k \cdot (L - 1)), \quad (13)$$

where L is the length of the raw sorted score vector.

3.4. Loss function

We employ the DICE loss [58] and the binary cross-entropy (BCE) loss [59] to measure the dissimilarity between the predicted forgery area and the ground truth (GT) mask in the model training stage. Additionally, we present an innovative loss function known as adaptive loss, which enables the model to optimize parameter values effectively and enhance detection accuracy throughout the training phase.

DICE loss. A commonly used metric to assess how similar two samples are to one another is the DICE coefficient. The function can be defined as:

$$\text{DICE} = \frac{2|X \cap Y|}{|X| + |Y|}, \quad (14)$$

where $|\cdot|$ denotes elements number, X denotes the given forged region mask. Y denotes the predicted forged region. In our model, we define the DICE loss as follows:

$$L_{\text{DICE}} = 1 - \frac{2 \cdot \sum_k (y_k \odot \hat{y}_k)}{\sum_k (y_k + \hat{y}_k)}, \quad (15)$$

where y represents the GT mask. $\hat{y}$ denotes the predicted mask for forgery areas. $\odot$ denotes element-wise multiplication. By minimizing the DICE loss, the proposed model can ensure the predicted forgery area is more similar to the GT.

BCE loss. We introduce the BCE loss to achieve stable training. Typically, when the positive samples are small objects, the DICE loss undergoes severe oscillations. When detecting images with only foreground and background, incorrect predictions of some pixels of small targets can result in a significant change in DICE loss and a sharp variation in gradients. BCE loss has great effectiveness in situations where the goal is to minimize the binary cross-entropy between the predicted value and the target value. The function of BCE can be represented as follows:

$$L_{\text{BCE}} = - \sum_{(i,j)} [\mathcal{G}(i, j) \log(\mathcal{T}(i, j)) + (1 - \mathcal{G}(i, j)) \log(1 - \mathcal{T}(i, j))]. \quad (16)$$

where $\mathcal{G}(i, j) \in [0, 1]$ denotes the original area predicted probability of pixel (i, j) , and $\mathcal{T}(i, j)$ denotes corresponding predicted probability to be tampered.

Adaptive loss. For adaptive attention, we introduce two learnable parameters $S1$ and $S2$ for Toto adaptively retain the attention weights. To stabilize the training, we add a regularization loss as follows:

$$L_{\text{Adapt}} = - \sum (\|S1\|_2 + \|S2\|_2). \quad (17)$$

where $\|\cdot\|_2$ denotes the ℓ_2 norm.

At last, the following formula represents the loss function of our model:

$$L_{\text{total}} = \alpha \cdot L_{\text{DICE}} + \beta \cdot L_{\text{BCE}} + \gamma \cdot L_{\text{Adapt}}. \quad (18)$$

where $\alpha, \beta, \gamma \in (0, 1)$ are weighting factors for corresponding loss function.

4. Experiments and comparisons

This section presents the experiment results of our model and compares them with state-of-the-art methods.

Table 1
Description of the Five Datasets Used in Our Experiments.

Dataset	Number of images			Image resolution
	Authentic	Copy-move	Total	
CASIA v1.0 [60]	800	459	1725	From 240×160 to 900×600
COVERAGE [61]	100	100	200	From 190×334 to 551×556
CoMoFoD [62]	0	200	200	512×512
MICC-F220 [63]	110	110	220	From 722×480 to 800×600
Our synthetic dataset	0	120 000	120 000	512×512

4.1. Datasets

Table 1 provides detailed information on the datasets used in research. We employ four publicly available datasets: CASIA v1.0 [60], COVERAGE [61], CoMoFoD [62], and MICC-F220 [63]. These datasets are chosen due to their frequent usage in detecting copy-move forgery tasks and they cover a broad spectrum of forgery scenarios. Particularly, the CoMoFoD dataset comprises 200 copy-move forgery images, each with a resolution of 512×512 . For each image, six distinct post-processing methods were applied, including brightness changing (BC), image blurring (IB), JPEG compression (JC), contrast adjusting (CA), noise addition (NA), and color reduction (CR). Moreover, the CASIA v1.0 dataset, which consists of 459 copy-move forgery images, underwent meticulous examination to eliminate mislabeled images and incorrect masks, ensuring the reliability of the experimental results.

To address the limitations of existing datasets, such as limited size and lack of labels and masks for tampered regions, we generated a comprehensive dataset for detection training and employed the synthetic dataset to pre-train our model. Inspired by the methodology proposed in [64], we adopt a similar approach and leverage the COCO [65] as a source of synthetic data. In addition, we use the ImageNet dataset as the source of our synthetic dataset [66]. We synthesize samples by selecting irregularly shaped regions in the image and pasting them into the original image. During the dataset synthesis process, random affine transformations are applied, with rotation angles uniformly distributed in the range $\mathbb{U}(-15^\circ, 15^\circ)$ and scale changes uniformly distributed in $\mathbb{U}(0.85, 1.15)$. Additionally, consistent translation alterations were applied across the entire image. To ensure non-overlapping source and destination regions, a trial-and-error approach is utilized to evaluate random affine transforms. Furthermore, Poisson blending is employed to seamlessly blend the source and target images, effectively eliminating the boundary between them and creating a more realistic appearance. To enhance the realism of the dataset, we also introduced Gaussian noise and applied random image compression algorithms to the generated data. The generated dataset provides the tampered area mask. To facilitate training, validation, and testing, the synthetic dataset is partitioned into sets with a ratio of 7:2:1.

4.2. Implementation details

Our copy-move forgery detection model is implemented using PyTorch. The backbone network employed in our model is ResNet50. During the training process of the TransCMFD model, we use the Adam optimizer with an initial learning rate of $1e-4$. For the uniformity of training samples, we adjust the size of the input image to 512×512 . To accelerate training, we leverage multi-GPU settings. The batch size of the training process is set to 16. The training process is implemented on a system comprising eight NVIDIA GeForce RTX 2080 Ti GPUs. These GPUs provide the computational resources required for efficient and parallelized training of the model.

4.3. Evaluation metrics

We evaluate the effectiveness of the compared model and our model using the F1 score and Intersection over Union (IoU) score, which are a

widely used metric for binary classification tasks. The precision metrics Pre and recall measurement Rec can be calculated as:

$$Pre = \frac{T_P}{T_P + F_P}, \text{ and } Rec = \frac{T_P}{T_P + F_N}, \quad (19)$$

where T_P denotes the number of forgery region pixels correctly predicted as forgery region pixels. F_P denotes the number of original region pixels incorrectly predicted as forgery region pixels. F_N denotes the number of forgery region pixels incorrectly predicted as original region pixels. Besides, to evaluate pixel-level performance, the IoU score is defined as follows:

$$IoU = \frac{T_P}{T_P + F_P + F_N}. \quad (20)$$

The precision measures how many positive samples were correctly predicted out of all the positive samples in the prediction results, reflecting the accuracy of the prediction. The recall, on the other hand, is defined as the ratio of genuine positive samples to all positive samples in the original dataset, measuring the completeness of the prediction. Neither precision nor recall can comprehensively evaluate the effectiveness of the model. So we also need to use F1, which combines precision and recall, to measure the final performance combined with IoU score. Specifically, the $F1$ score is defined as:

$$F1 = 2 \cdot \frac{Pre \cdot Rec}{Pre + Rec}. \quad (21)$$

4.4. Experiment results

We assess our model's performance by comparing the model with some baseline models, which comprise both traditional and several deep learning-based models. Traditional algorithms used for comparison are the keypoint-based algorithm zandi with iterative interest points [67] and block-based dense field method cozzolino [68]. In addition to traditional algorithms, we also selected deep learning algorithms as baseline models. These deep learning models evaluated for comparison are busternet [26], CMSDNet [13], doa-gan [40], UCMNet [69], and CMCNet [70].

We also introduce an additional model, the transformer copy-move forgery detection (transcmfd) model. The model utilizes only an encoder-decoder architecture, a traditional transformer, and a feature similarity module. By including this model, The reason we chose this model is to reflect the superior performance of our proposed adaptive transformer module.

4.4.1. Results on existing datasets

To demonstrate the superior performance of our model, we conducted quantitative experiments on four benchmarks: MICC-F220, COVERAGE, CoMoFoD, and CASIA v1.0, as described in Table 1. Since the MICC-F220 does not provide the GT mask, we use a confusion matrix to evaluate the performance of MICC-F220 at the image level instead of the pixel-level performance test. In the confusion matrix, we mainly use six indicators: True Positives (TP), False Positives (FP), True Negatives (TN), False Negatives (FN), True Positive Rate (TPR), and False Positive Rate (FPR). On the contrary, pixel-level performance tests are carried out on CASIA v1.0, CoMoFoD, and COVERAGE. TPR and FPR can be computed as follows:

$$TPR = \frac{TP}{TP + FN}, \text{ and } FPR = \frac{FP}{TN + FP}, \quad (22)$$

Table 2

Comparison of the pixel-level F1 and IoU scores for different copy-move forgery detection methods on the selected datasets.

Models	synthetic dataset		CASIA v1.0		CoMoFoD		COVERAGE	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU
zandi [67]	0.1839	0.1061	0.4761	0.3244	0.3796	0.2533	0.4235	0.2683
cozzolino [68]	0.1045	0.0562	0.2543	0.1459	0.4440	0.2933	0.6480	0.4791
busternet [26]	0.4761	0.3124	0.4582	0.2973	0.4926	0.3297	0.6180	0.4311
CMSDNet [13]	0.5032	0.3368	0.5380	0.3678	0.5110	0.3436	0.6770	0.5219
doa-gan [40]	0.5549	0.3862	0.4144	0.2653	0.5507	0.3814	0.5590	0.3872
UCMNet [69]	0.6125	0.4426	0.5831	0.4117	0.5820	0.4103	0.6433	0.4769
CMCFNet [70]	0.6332	0.4632	0.5712	0.4061	0.5233	0.3552	0.7168	0.5592
transcmfd	0.6907	0.5291	0.5540	0.3838	0.5833	0.4113	0.6953	0.5332
Ours	0.7356	0.5815	0.5867	0.4155	0.6320	0.4622	0.7248	0.5674

Table 3

Comparison of the image-level performance using confusion matrix for different copy-move forgery detection methods on the MICC-F220.

Models	TP	FP	TN	FN	TPR(%)	FPR(%)
zandi [67]	54	71	19	76	41.54	78.89
cozzolino [68]	103	27	77	13	88.79	25.96
busternet [26]	63	30	75	52	54.78	28.57
CMSDNet [13]	59	36	76	49	54.63	32.14
doa-gan [40]	58	33	82	47	55.24	28.70
UCMNet [69]	61	39	79	41	59.80	33.05
CMCFNet [70]	61	35	82	42	59.22	29.91
transcmfd	63	31	80	46	57.80	27.93
Ours	64	26	90	40	61.54	22.41

The average pixel-level F1 and IoU scores of the baseline algorithms and the proposed model on the CASIA v1.0, CoMoFoD, and COVERAGE are shown in Table 2. We consistently get the best results on the CASIA v1.0, CoMoFoD, and COVERAGE tests with our model, outperforming the doa-gan and CMSDNet, demonstrating competitive performance in copy-move forgery detection. Specifically, over the busternet, our model improves the F1 on average by 0.1249, and improves the IoU score on average by 0.3719. On the CASIA v1.0, CoMoFoD, and COVERAGE, our model can improve an average F1 by 0.0725 over the CMSDNet, and improves the IoU score on average by 0.1873. On the CASIA v1.0, CoMoFoD, and COVERAGE, our model can improve an average F1 by 0.0729 over the UCMNet and improves the IoU score on average by 0.1085. On the CASIA v1.0, CoMoFoD, and COVERAGE, our model can improve an average F1 by 0.0820 over the CMCFNet and improve the IoU score on average by 0.1130. Moreover, the proposed algorithm improves an average F1 of 0.1398 over the doa-gan and 0.0369 over the transcmfd on the COVERAGE, CoMoFoD, and CASIA v1.0. We also noticed that the two traditional methods performed well on the COVERAGE. The primary cause is that no special processing, like rotation, was applied to the images in the COVERAGE dataset. Since the COVERAGE dataset is relatively small, traditional methods can still be effective in cases of limited data. Traditional methods often employ hand-designed feature extractors, which can better adapt to situations with limited data and mitigate problems such as overfitting that may arise with deep learning models on small datasets.

Table 3 presents the confusion matrix results of the baseline algorithms and the proposed model on MICC-F220. Our method performs superior on the MICC-F220 dataset compared to all deep learning methods on confusion matrix results. Our method achieves 61.54% in TPR, indicating that the model performs well in identifying positive samples. Meanwhile, the FPR is only 22.41%, which is lower than other comparison methods. This means the model is better than other methods in reducing false positives. However, the cozzolino method outperforms all deep learning models on the TPR metric. This may be due to the lower complexity and noise present in the MICC-F220 dataset, where traditional methods tend to perform better. Deep learning models require many samples to learn meaningful features. In cases where the dataset is small, traditional methods may adapt better to the

data due to their lower sample requirements. However, it is noteworthy that even with these difficulties, our model outperforms all other deep learning models.

4.4.2. Results on our synthetic dataset

During our evaluation of synthetic datasets, 70% is training data, 20% is validation data, and the remainder data is test dataset, as explained in Section 4.1. We report the average pixel-level F1 and IoU score results for 1,200 test samples in Table 2.

The following are our main conclusions from the experiment: (a) Deep learning-based approaches, including the busternet, CMSDNet, doa-gan, UCMNet, CMCFNet, transcmfd, and the proposed model, outperform traditional methods. Deep learning-based models can learn high-level representations from raw data through multi-layer nonlinear transformations. By leveraging this capability, deep learning models can capture more complex and abstract features, which better describe the essential characteristics of the data. (b) Among the deep learning-based methods, our model demonstrates the best performance. This can primarily be attributed to the integration of the adaptive transformer, which effectively improves the forgery detection accuracy. The adaptive transformer and CNN models leverage complementary information. The adaptive transformer focuses on capturing global features, while the CNN focuses on extracting more local features within the image. This combination allows our model to leverage global and local information, enhancing its ability to accurately detect copy-move forgery.

4.4.3. Visualization results

The visualization results of selected models on a selected portion of the image are shown in Fig. 3. Our model shows better performance than other models in generating accurate masks for tampered areas, demonstrating its precise localization capability during deep matching. However, it is also noticeable that our model struggles to locate the original region. This highlights the need for further research to investigate the differentiation of genuine regions from copy-move forgery regions that appear similar.

4.4.4. Robustness analysis

We perform robustness analysis experiments on the CoMoFoD dataset. CoMoFoD has 200 copy-move images with six different attack techniques. Each image has a resolution of 512×512 . These six augmentation processes include BC, CA, CR, IB, JC, and NA. We employ a BC attack within the range of (0.01, 0.95) and (0.01, 0.90) to generate visually imperceptible images after the attack. We perform CA on the input copy-move forgery image by adjusting the intensity value of the image to a new range. This process effectively reduces the output image intensity range. The choice of lower and upper bounds is motivated by the observation that CA within the (0.01, 0.95) and (0.01, 0.90) ranges have a minimal visual impact on tampered images, while adjustments within the range of (0.01, 0.8) produce more noticeable changes. We perform the CR attack by adjusting the intensity weights of the tampered image. Specifically, we reduce the number of intensity

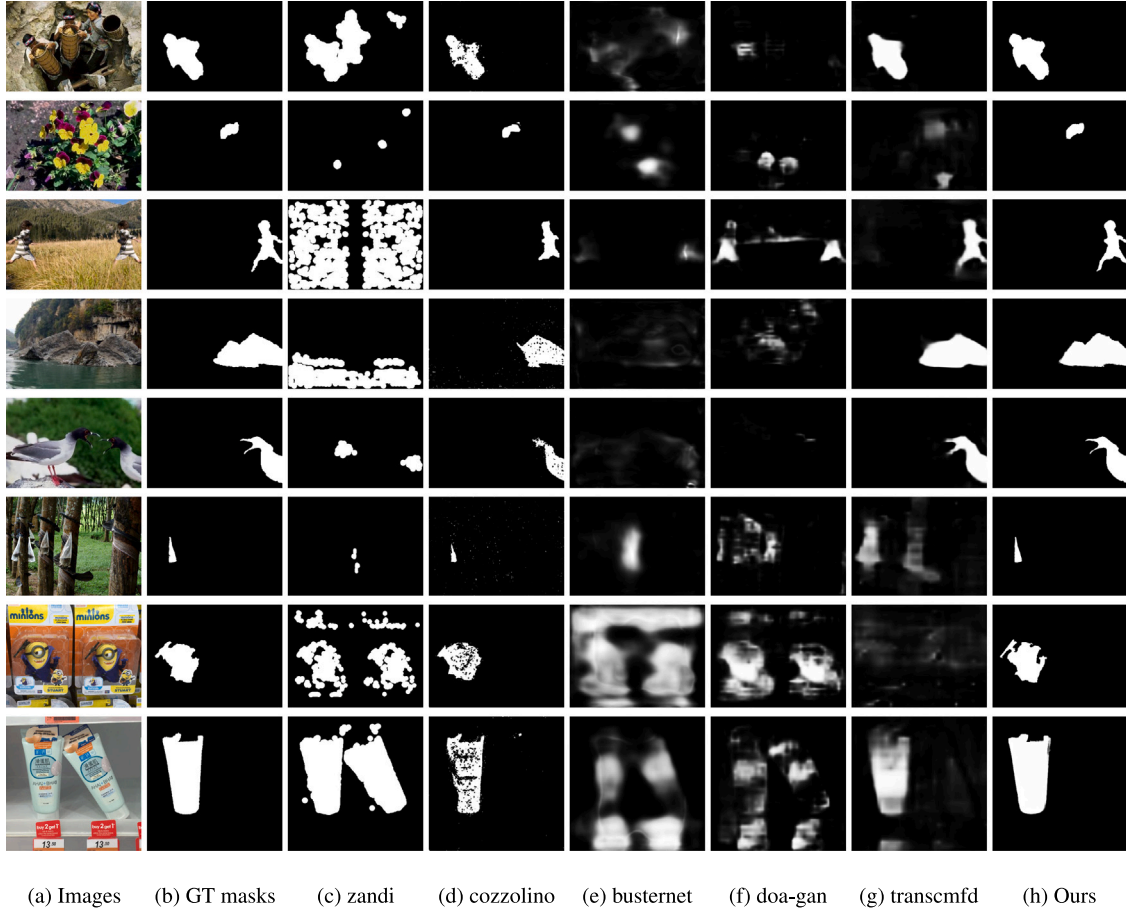


Fig. 3. Prediction visual results generated by various copy-move forgery detection methods.

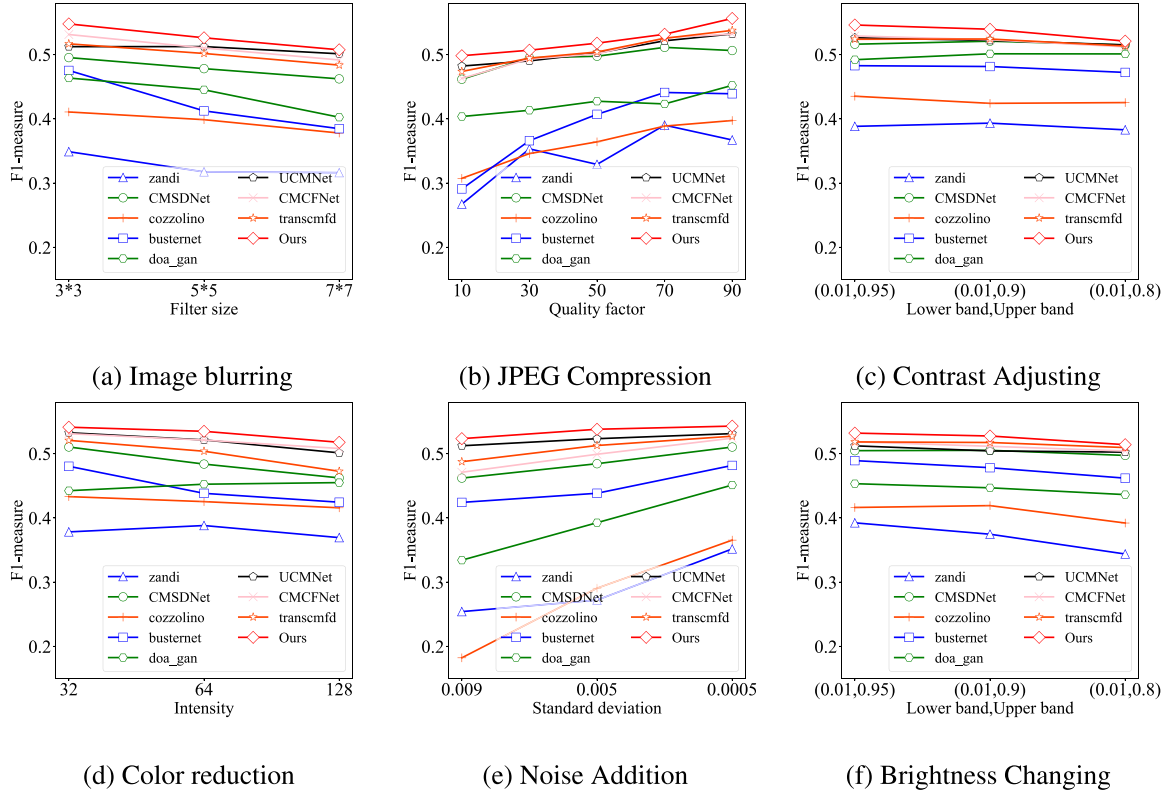


Fig. 4. Pixel-level F1 of our model under different attacks. The used dataset is CoMoFoD.

Table 4

Configurations of ablation experiment networks, E-D means CNN encoder-decoder, ConvBlock denotes convolution block and FSM means Feature similarity module.

	E-D	ConvBlock	Transformer	Adaptive transformer	FSM
Network#1	✓				
Network#2	✓		✓		
Network#3	✓				✓
Network#4	✓	✓			✓
Network#5	✓		✓		✓
Network#6	✓			✓	✓

Table 5

Comparison of pixel-level F1 and IoU scores on three datasets in ablation experiments.

Ablation network	CASIA v1.0		CoMoFoD		COVERAGE	
	F1	IoU	F1	IoU	F1	IoU
Network#1	0.4003	0.2507	0.4812	0.3169	0.4078	0.2561
Network#2	0.5267	0.3561	0.5113	0.3432	0.5736	0.4009
Network#3	0.5337	0.3622	0.5104	0.3426	0.6080	0.4213
Network#4	0.5212	0.3526	0.5311	0.3531	0.6124	0.4311
Network#5	0.5540	0.3826	0.5833	0.4121	0.6953	0.5314
Network#6	0.5867	0.4153	0.6320	0.4612	0.7248	0.5677

levels for each color channel of the manipulated image from 256 to 32, 64, and 128 levels, respectively. Tampered images under CR are visually indistinguishable from the original tampered images. The IB attack on tampered images is achieved by convolving the original image with a 3×3 , 5×5 , and 7×7 average filter. Images under 7×7 filtering attacks will appear visually blurry. For the JC attack, we select the JPEG quality factor to range from 10 to 90. During copy-move forgery image saving, the image resolution remains unchanged. This choice of settings is motivated by the observation that for JPEG forgery images with a quality factor lower than 70, visual degradation and block artifacts are noticeable, while for quality factors higher than 70, the visual changes are negligible. In the NA attack, we add different Gaussian noises to the original tampered image. The noise variance values we chose are 0.009, 0.005, and 0.0005, respectively. The noise we add to the tampered image has a mean value of 0.

The experiments aim to estimate the robustness of our model by subjecting the images to varying degrees of attack. The results are illustrated in Fig. 4 display the pixel-level F1 results for various methods. The values are averaged over 200 samples. The F1 result is shown on the y-axis in each subfigure, and various attack parameters are shown on the x-axis. It is evident that both the baseline algorithms and our model experience a decrease in both the F1 scores for each type of attack from Fig. 4. However, our model consistently achieves the highest performance across all six levels of attack. Notably, it can be found that deep learning-based models exhibit stronger resistance to attacks compared to traditional methods.

To verify the impact of various components in the model on the detection performance of our model, we used various networks in ablation experiments. Generally, the performance of our forgery detection model is influenced by three components: encoder-decoder structure, adaptive transformer, and feature similarity module. The encoder-decoder facilitates automatic feature extraction and learning. The adaptive transformer can adaptively learn the effective features of the tampering area and improve the quality of representation, and it can better capture the difference between tampered and original regions than the original transformer. The feature similarity module can accurately measure the similarity between the original region and the synthesized region.

We designed six distinct models for ablation experiments, as presented in Table 4. Network#1 represents the pure encoder-decoder structure network. Network#2 extends the encoder-decoder architecture by incorporating the common transformer encoder module. Network#3 incorporates the feature similarity module into the encoder-decoder structure. Network#4 combines the encoder-decoder structure, convolution block replacement for transformer, and feature similarity module. Network#5 combines the encoder-decoder structure,

Table 6

Confusion matrix results of MICC-F220 for copy-move forgery detection methods in ablation experiments.

Ablation network	TP	FP	TN	FN	TPR(%)	FPR(%)
Network#1	47	85	48	40	54.02	63.91
Network#2	53	46	79	42	55.79	36.80
Network#3	58	31	86	45	56.31	26.50
Network#4	57	40	81	42	57.58	33.60
Network#5	63	31	80	46	57.80	27.93
Network#6	64	26	90	40	61.54	22.41

transformer module, and feature similarity module. Lastly, Network#6 includes the encoder-decoder structure, proposed adaptive transformer module, and feature similarity module. Through different combinations of these components, we can discover the impact of these components on forgery detection performance from the results of these networks.

The MICC-F220 dataset lacks GT masks for the forgery regions. The proposed model's ablation experiment performance is assessed with the confusion matrix at the image-level, as shown in Table 6. Subsequently, we conduct pixel-level ablation experiment evaluations using the CASIA v1.0, CoMoFoD, and COVERAGE datasets. The corresponding F1 and IoU scores are presented in Table 5. Several inferences can be made based on these experimental findings:

- Network#6 achieves the highest average pixel-level F1 and IoU scores across the CASIA v1.0, CoMoFoD, and COVERAGE, demonstrating superior localization performance. Compared to Network#3, both Network#5 and Network#6 have improved significantly on the pixel-level F1 and IoU scores. Network#5 improves by an average of 6.02% over Network#3 across the CASIA v1.0, CoMoFoD, and COVERAGE. Network#6 improves by an average of 9.71% over Network#3 across the CASIA v1.0, CoMoFoD, and COVERAGE. Network#5 improves F1 by an average of 9.89% and IoU score by an average of 16.2% over Network#4 across the CASIA v1.0, CoMoFoD, and COVERAGE. Network#6 improves F1 by an average of 16.6% and IoU score by an average of 26.7% over Network#4 across the CASIA v1.0, CoMoFoD, and COVERAGE. This shows that the transformer and adaptive transformer can improve the detection accuracy more when combined with the overall module compared to the traditional convolution module. The result of Network#6 is also significantly improved compared to Network#5, emphasizing the importance of incorporating the proposed transformer module in the localization of the forgery area.
- In comparison to Network#1, the pixel-level F1 and IoU scores for the CASIA v1.0, CoMoFoD, and COVERAGE improve with the addition of the feature similarity module in Network#3. This suggests that the feature similarity module plays a substantial position in enhancing the performance of the TransCMFD.
- When compared to the other models of ablation experiments, the Network#6, which combines the feature similarity module with the adaptive transformer module, exhibits notable improvements in pixel-level F1 and IoU scores across the CASIA v1.0, CoMoFoD, and COVERAGE. These findings demonstrate that modules of the TransCMFD can enhance the tampered region detection performance.

The ablation experiment results of TransCMFD on the MICC-F220 achieve notable improvements, as illustrated in Table 6. Network#6 performs excellently in the ablation experiment on the MICC-F220. The TPR of Network#6 is 61.54%, which is higher than other models. This means that Network#6 has higher accuracy in detecting tampered areas. The FPR of Network#6 is 22.41%, which is significantly lower than other network models. This indicates that Network#6 performs well in reducing the false positive rate. In conclusion, these experiment results demonstrate that the proposed module is effective for copy-move forgery detection on MICC-F220.

5. Conclusion

In this paper, we propose a novel adaptive transformer deep learning model named TransCMFD for copy-move forgery detection. TransCMFD can effectively utilize tampered area features by combining a novel adaptive transformer module with CNN. By utilizing both global and local low-level convolutional features, our model can accurately localize and extract the features from the tampered regions. Compared with traditional manual feature extraction methods and other deep learning-based methods, our model has great flexibility to adapt to different tasks and datasets. This flexibility enhances the model's generalization ability and accuracy. Extensive experiments demonstrate that our model performs outstanding pixel-level performance on the CASIA v1.0, COVERAGE, and CoMoFoD datasets, and performs better on the MICC-F220 dataset using confusion matrix for image-level evaluation on the MICC-F220 dataset. Compared with baseline methods, our model exhibits high accuracy in locating forgery regions and remains robust against post-processing effects, including noise, color reduction, brightness adjustment, blurring, and JPEG compression.

CRedit authorship contribution statement

Enji Liang: Writing – original draft, Visualization, Software, Methodology, Data curation, Conceptualization. **Kuiyuan Zhang:** Writing – review & editing, Validation, Formal analysis, Conceptualization. **Zhongyun Hua:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Yuanman Li:** Writing – review & editing, Formal analysis, Conceptualization. **Xiaohua Jia:** Writing – review & editing, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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Enji Liang received his B.E. degree in information engineering from Jilin University, Changchun, China. He is currently pursuing the Ph.D. degree in computer science and technology with the School of Computer Science and Technology, Harbin Institute of Technology, Shenzhen. His current research interests include multimedia forensics, computer vision, and deep learning.

Kuiyuan Zhang received his M.S. degree in Computer Science and Technology from Harbin Institute of Technology, Shenzhen, China. He is currently pursuing the Ph.D degree in Computer Science and Technology from Harbin Institute of Technology, Shenzhen, China. His current research interests include image compressive sensing, image encryption and multimedia deepfake detection.

Zhongyun Hua (Senior Member, IEEE) received the B.S. degree in software engineering from Chongqing University, Chongqing, China, in 2011, and the M.S. and Ph.D. degrees in software engineering from the University of Macau, Macau, China, in 2013 and 2016, respectively. He is currently a Professor with the School of Computer Science and Technology, Harbin Institute of Technology, Shenzhen, China. His works have appeared in prestigious venues such as USENIX Security, ACM CCS, ICML, IEEE TDSC, IEEE TIFS, and IEEE TSP. He has been recognized as a 'Highly Cited Researcher 2024'. His current research interests are focused on Applied Cryptography, Multimedia Security, Trustworthy AI, and Nonlinear Systems and Applications. He has published more than 120 papers on the subject, receiving more than 8500 citations.

Yuanman Li (Member, IEEE) received the B.Eng. degree in software engineering from Chongqing University, Chongqing, China, in 2012, and the Ph.D. degree in computer science from the University of Macau, Macau, in 2018. From 2018 to 2019, he was a Post-Doctoral Fellow with the State Key Laboratory of Internet of Things for Smart City, University of Macau. He is currently an Associate Professor with the College of Electronics and Information Engineering, Shenzhen University, Shenzhen, China. His current research interests include data representation, multimedia security and forensics, computer vision, and machine learning.

Xiaohua Jia (F'13) received the B.Sc. and MEng degrees in 1984 and 1987, respectively, from the University of Science and Technology of China, and the DSc degree in 1991 in information science from the University of Tokyo. He is an adjunct in Harbin Institute of Technology (Shenzhen) while performing this work. He is currently the chair professor with the Department of Computer Science at the City University of Hong Kong. His research interests include cloud computing and distributed systems, computer networks, wireless sensor networks and mobile wireless networks. He is an editor of the IEEE Transactions on Parallel and Distributed Systems (2006–2009), Wireless Networks, Journal of World Wide Web, Journal of Combinatorial Optimization, etc. He is the general chair of ACM MobiHoc 2008, TPC co-chair of IEEE MASS 2009, area-chair of IEEE INFOCOM 2010, TPC co-chair of IEEE GlobeCom 2010-Ad Hoc and Sensor Networking Symposium, and Panel co-chair of IEEE INFOCOM 2011. He is fellow of the IEEE Computer Society.